

NON-ABELIAN CHARGED VORTICES AS COSMIC STRINGS

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Received 1 July 1985; revised manuscript received 11 February 1986

We study vortex solutions in non-abelian gauge theories with spontaneous symmetry breaking. We construct string configurations with electric charge showing that they necessarily have infinite energy per unit length. Although this prevents its application in problems like catalysis of proton decay, we show, by analyzing the cosmological evolution of the strings, that they can produce density fluctuations leading to galaxy formation.

Extended solutions of field theory classical equations of motion have been playing a role of increasing importance in particle physics^{‡1} and, more recently, in cosmology [2].

Concerning gauge theories with Higgs fields, vortex and monopole solutions have been investigated both in their classical and quantum aspects, their topological properties being at the origin of rather unexpected and important phenomena. In particular, Rubakov [3] and Callan [4], studying dyon excitations, discovered that magnetic monopoles can serve as catalyzer of proton decay.

Because of their very large mass, the monopoles of the grand unified theories (GUTs) could be only of cosmological origin. Generated at a phase transition in the early history of the universe [5], their relevance in its evolution has been analyzed in many investigations [2,5].

It is well known that topological vortices also appear in GUTs, their cosmological effects having been revealed by recent works [5–12]. Generated at the grand-unification phase transition, they may provide a natural origin for the density fluctuations required to initiate the condensation process leading to galaxy formation. It has been also conjectured that GUTs vortices, in analogy with monopoles, may catalyze proton decay [13].

We examine in this work some properties of non-

abelian vortices in connection with the phenomena referred above. We discuss in particular the possibility of existence of vortices with dyonic charge. We show that charged vortices (here “charge” cannot be taken in the usual sense since the existence of a vortex solution require all vector mesons to be massive, see below) necessarily imply a divergent energy per unit length. This result constitutes a serious limitation in the uses of these solutions in semiclassical investigations like those related to proton decay. On the contrary, they do not introduce any difficulty in a cosmological context, as first noted by Vilenkin and Everett [9] in their studies of global strings. In view of this fact, we also investigate the relevance of our charged vortices in connection with density perturbations needed to initiate the formation of galaxies.

Although the arguments that follow are rather general and apply to “realistic” theories, such as the SU(5) grand unified theory, it is useful to start by considering the simplest non-abelian theory that has vortex solutions [14–18]. This is the theory of a gauge field A_μ taking values in the Lie algebra of $G = \text{SU}(2)$, coupled to two Higgs fields in the adjoint representation, with lagrangian density

$$L = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} D_\mu \Phi D^\mu \Phi + \frac{1}{2} D_\mu \Psi D^\mu \Psi - V(\Phi, \Psi), \quad (1)$$

with

$$D_\mu \Phi = \partial_\mu \Phi + e A_\mu \wedge \Phi, \quad (2)$$

^{‡1} For a complete list of references see e.g. ref. [1].

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + e A_\mu \wedge A_\nu, \quad (2 \text{ cont'd})$$

where ϕ, ψ, A_μ stands for isovectors.

As is well known, one has to introduce at least two scalar multiplets for a complete symmetry breaking ensuring topological stability of the vortex solution. For example, one may take the potential in the form

$$V(\phi, \psi) = \frac{1}{8} g^2 (\phi \cdot \phi - \eta^2)^2 + \frac{1}{8} g'^2 (\psi \cdot \psi - \eta'^2)^2, \quad (3)$$

choosing the vacuum solutions ϕ_0 and ψ_0 not parallel in isospace. In this way, the relevant homotopy group, $\pi_1(G/H) = Z_2$, since the subgroup H , leaving invariant the vacuum reduces to the center of $SU(2)$. We then can have, apart from the trivial vacuum, one topologically stable vortex solution. An appropriate Ansatz for a purely magnetic, static, axially symmetric vortex solution is the following:

$$\phi = f(\rho) \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix}, \quad \psi = \eta' \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad (4)$$

and

$$A_\theta^a = H(\rho) \delta^{a3} \quad (5)$$

with all the other A_μ components being zero. The boundary conditions are:

$$f(0) = H(0) = 0,$$

$$f(\rho) = \eta, \quad H(\rho) = -1/e\rho, \quad \text{for } \rho \rightarrow \infty. \quad (6)$$

Using eqs. (4), (5), the angular dependence is trivially separated and the resulting non-linear radial equation can be solved numerically. The solution corresponds to a gauge-field tensor decreasing with a characteristic length $\lambda = 1/m_V$, with the vector meson mass m_V given by the expression

$$m_V^2 = e^2(\eta^2 + \eta'^2), \quad (7)$$

and a Higgs field ϕ increasing with characteristic length $\xi = 1/m_S$, where the scalar mass m_S is

$$m_S^2 = g^2 \eta^2 \quad (8)$$

from zero at the origin to its vacuum value at infinity. This corresponds to a string configuration of width ξ and penetration depth λ . If ξ and λ are of the same order of magnitude ($m_V \sim m_S$), we have a well defined vortex line or a well defined string.

Let us try now to get a vortex configuration with non-zero "electric field". Of course, the term "electric" has to be qualified, since in the model we are considering all vector mesons are massive and hence there is no electric charge in the usual sense. However, as explained in ref. [19] one can connect such a charge with angular momentum in the usual way.

Adopting the $A_0 = 0$ gauge, we can study dyon excitations by introducing a time-dependent gauge rotation of the vortex configuration defined in eqs. (4), (5):

$$A_i^\Lambda = U_\Lambda A_i U_\Lambda^{-1} + (i/e) U_\Lambda \partial_i U_\Lambda^{-1}, \quad (9)$$

$$U_\Lambda = \exp[i t_3 \Lambda(\rho, t)], \quad (10)$$

where $A_i \equiv A_i^a t^a$ and $t^a = \sigma^a/2$ are the $SU(2)$ generators. We have chosen Λ so as to maintain axial-symmetry. Being the rotation parallel to the ψ direction, only ϕ is transformed:

$$\phi^\Lambda = U_\Lambda \phi U_\Lambda^{-1}, \quad \phi = \phi^a t^a. \quad (11)$$

Associated with transformation (10), we can define an electric charge density which takes the form

$$J_0 = \partial^i F_{i0}^3. \quad (12)$$

Had we considered instead of (10) a rotation parallel to the ϕ direction, the associated charge density would read

$$J_0^\phi = (1/\eta) \partial^i (F_{i0} \cdot \phi), \quad (13)$$

but the resulting charge is the same.

Using eqs. (5), (9), (13) we get

$$J_0 = (1/e) \rho^{-1} (\partial/\partial \rho) [\rho \partial^2 \Lambda(\rho, t) / \partial \rho \partial t]. \quad (14)$$

Integration of (14) leads to the following expression for the charge per unit length q :

$$q \equiv \int \rho \, d\rho \, d\theta \, J_0 = \frac{2\pi}{e} \left(\rho \frac{\partial^2 \Lambda}{\partial \rho \partial t} \right) \Big|_0^\infty \quad (15)$$

We then see that in order to have $q \neq 0$, $\partial \Lambda / \partial t \sim \log \rho$ either at the origin or at infinity.

Suppose one makes the choice

$$\partial \Lambda(0, t) / \partial t = 0,$$

$$\partial \Lambda(\rho, t) / \partial t \rightarrow (1/2\pi) e K \log \rho, \quad \text{for } \rho \rightarrow \infty, \quad (16)$$

with K an arbitrary constant. Then, the charge per unit length becomes

$$q = K. \quad (17)$$

At the classical level this charge is completely unrestricted and only after a detailed quantum analysis one can decide whether the total charge carried by the vortex is an integer multiple of the fundamental charge e (as it is the case for the monopole).

As we pointed above our solution corresponds to the $A_0 = 0$ gauge. One can extend however the ansatz (4)–(6) to get an analogous solution in a $A_0 \neq 0$ gauge by writing, for example $A_0^a = A(\rho)\delta^{a3}$ as an ansatz for the scalar potential. Again the equations of motion are separable into radial and angular parts and one discovers that in order to have a charged solution $A(\rho)$ has to behave as $\log \rho$ at the origin, the same kind of behavior encountered for $\partial\Lambda/\partial t$, eq. (15).

Now, it is evident that the behavior (16) makes the string energy per unit length divergent, since the solution corresponds to a radial electric field $E_\rho \sim \rho^{-1}$ at infinity. This invalidates the use of a charged vortex as a soliton configuration in semiclassical analysis. In particular, the investigation à la Callan–Rubakov of a fermion system in the presence of dyons cannot be naively extended to the case of charged vortices, as in ref. [13].

There is however a field where this kind of divergent-energy solutions, usually discarded in particle-physics applications based on solitons, do not introduce any difficulty. On the contrary, as first stressed by Vilenkin and Everett [9], they may be welcome in a cosmological context where the occurrence of topologically stable strings in some GUTs may be a positive feature. Indeed, as pointed out by Zel'dovich [6], strings formed at the grand-unification phase transition could later provide the density perturbations necessary to seed the process of galaxy formation.

The analysis of the cosmological evolution of strings in models with spontaneous breaking of a local symmetry, developed by Vilenkin [7] has been extended to the case of global symmetries in simple models [9] (like the Goldstone model) showing that infinite energy density strings are also present in this case. However, by noting that the scale L of strings formed at the phase transition (that is, the typical curvature radius of the strings and the typical distance between adjacent strings are $\sim L$) provides a natural cutoff, one can get a finite answer.

In view of this fact, let us compute the string energy density using this cutoff. It is convenient to define dimensionless variables:

$$x = m_V \rho, \quad \tilde{f}(\rho) = f/m_V, \quad \tilde{\eta} = \eta/m_V, \quad (18)$$

in terms of which the energy density μ (the linear mass density of the string) is:

$$\mu = 2\pi m_V^2 \int_0^\infty T_{00}(x) x \, dx, \quad (19)$$

where T_{00} is given by

$$T_{00} = \frac{1}{2} [H'^2/x^2 + \tilde{f}'^2 + (1/x^2)\tilde{f}^2(1 + eH)^2 + \frac{1}{4}g^2(\tilde{f}^2 - \tilde{\eta}^2)] + \frac{1}{2}[\tilde{f}^2\dot{\Lambda}^2 + (1/e^2)\dot{\Lambda}'^2], \quad (20)$$

with

$$\dot{\Lambda} \equiv (1/m_V)\partial\Lambda/\partial t.$$

The Λ -independent terms in (20) just correspond to the energy density for the usual (purely magnetic) vortex. The other two terms have to be treated carefully, by using the cutoff L :

$$\begin{aligned} \mu &= \mu_{\text{mag}} \\ &+ \pi m_V^2 \int_1^{L/\xi} [\tilde{\eta}^2(eK/2\pi)^2 \log x \\ &+ (K^2/4\pi^2)x^{-2}] x \, dx, \end{aligned} \quad (21)$$

leading to

$$\begin{aligned} \mu &= \mu_{\text{mag}} + (K^2/4\pi) \{ (L^2/2\xi^2) [\log^2(L/\xi) \\ &- \log(L/\xi) + \frac{1}{2}] - \frac{1}{4} + \log(L/\xi) \}. \end{aligned} \quad (22)$$

The energy density for the purely magnetic part can be written in the form

$$\mu_{\text{mag}} = \eta^2 m(g^2/e^2), \quad (23)$$

where m is a slowly varying function which can be computed numerically [1]. It has been calculated exactly for the limiting case between type I and type II superconductivity, which corresponds to $e^2 = g^2$ [20]:

$$m(g^2/e^2 = 1) = \pi. \quad (24)$$

For grand-unified strings one has $\eta^2 \sim m_V^2/\alpha$ where α is the GUT-fine-structure constant $\alpha = e^2/4$, $\alpha \sim 0.02$.

Considering the dominant term in (22) we then have

$$\mu = \alpha \eta^2 Q^2 \log^2 L m_V, \quad (25)$$

where we have defined Q , the total charge of the vortex as

$$Q = KL. \quad (26)$$

As explained in ref. [9] the scale L of the system of strings at cosmic time t is $\sim t$ (if one considers that strings form closed loops, either because they started off that way or because they generate loops through self-interactions, L becomes the size of the loops). It has been found that the ratio of the contributions of strings to the total mass density is [6]:

$$\rho_s/\rho \simeq G\mu, \quad (27)$$

where ρ_s is the energy density due to strings, ρ the energy density of matter and G the gravitational constant.

Concerning closed loops, Vilenkin [7] suggested more recently that they have very long lifetimes contributing much more than do open strings to the total mass density. He has found

$$\rho_{\text{loops}}/\rho \simeq (G\mu)^{1/2}. \quad (28)$$

(Recently Kibble [12] has put on a firmer footing estimates on the number of loops, confirming (27), (28).)

Now, for charged vortices, using $m_V = 10^{15}$ GeV, we get

$$G\mu = G\alpha \eta^2 Q^2 \log^2 L m_V = 10^{-5} Q^2. \quad (29)$$

Since the order of magnitude of the density contrast [defined by eqs. (27), (28)] needed to trigger galaxy formation is of the order 10^{-3} to 10^{-5} , we see that strings with charge $Q \sim 1$ can produce adequate density fluctuations.

To summarize, we have shown that in order to have non-abelian charged vortices as classical solutions of gauge theories with spontaneously broken symmetry breaking, the price to be paid is an infinite energy per unit length. This makes it not possible to extend naively the studies of fermions in the presence of magnetic monopoles in order to analyze dyonic vortex excitations and the catalysis of proton decay. The

charged vortices can be of relevance, however, in a cosmological context and we have shown, following the ideas of Vilenkin and Everett [9], that such solutions can produce density fluctuations sufficiently large to explain the galaxy formation.

C.M.N. is supported by CONICET, Argentina and F.A.S. by C.I.C. Buenos Aires, Argentina. We also acknowledge support from the Deutsche Forschungsgemeinschaft (DFG).

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